

Setup Guide B

GitHub Codespaces

MLOps & Model Deployment

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Setup Guide B — GitHub Codespaces (recommended)

A Codespace is a full Linux computer that runs in your browser. Python, MLflow, Docker and git are already installed and configured. Nothing touches your laptop.

What you need: a browser (Chrome or Edge) and a **free GitHub account**. That's it. **Time to be up and running:** about **3 minutes**.

Before Day 1 — the only homework

1. Create a **free GitHub account** at **github.com** — and **click the verification link** in the confirmation email. An unverified account cannot create Codespaces.
2. Create a **free Hugging Face account** at **huggingface.co** — also **verify the email**. (Needed for Lab 5 on Day 3.)

Do this at home, not in class. Email verification on a busy campus network is the single most common reason a student is stuck at minute 10 of Day 1.

Step 1 — Open the project

Go to: github.com/anandvenkat4/mlops-sample-project

Step 2 — Create your Codespace

1. Click the green **Code** ▾ button.
2. Choose the **Codespaces** tab (not "Local").
3. Click **Create codespace on main**.

A new browser tab opens with VS Code in it and starts building. **Leave it alone** while it works.

Step 3 — Wait about 3 minutes

The first build installs everything and **trains a model for you automatically**. You'll see log output scrolling. This only happens once — resuming later takes seconds.

Step 4 — Verify it worked ✓

When the terminal is ready, type:

```
cd /workspaces/mlops-sample-project
bash .devcontainer/check.sh
```

You should see all four reported OK:

Check	Meaning
Packages	scikit-learn, MLflow, FastAPI etc. installed
Dataset	<code>data/loan.csv</code> exists
Model	<code>model.joblib</code> exists — setup trained it for you
Docker	Docker is running inside the Codespace

Optional extra proof:

```
ls model.joblib data/loan.csv      # both should be listed
docker run hello-world             # prints "Hello from Docker!"
```

All OK? You're ready for all three days.

The two things that confuse everyone

1. There is no virtual environment

Every lab worksheet shows two blocks. Use **"In Codespaces"** and **skip** `source .venv/bin/activate` — it will fail, and you don't need it. Packages are installed globally inside the container.

Your project is **always** at `/workspaces/mlops-sample-project`.

2. `localhost` in your own browser will not work

Your laptop and the Codespace are different computers.

What you're doing	In Codespaces
<code>curl http://127.0.0.1:8000/health</code> in the terminal	✓ Works exactly as written — the terminal <i>is</i> inside the Codespace
Opening a UI in your browser	✗ <code>localhost:8000</code> is <i>your laptop</i> . Use the PORTS tab or the pop-up

To open a running service:

1. Start it (e.g. `mlflow ui --backend-store-uri sqlite:///mlflow.db --port 5000`).
2. A pop-up appears bottom-right → click **Open in Browser**.
3. Missed the pop-up? Click the **PORTS** tab next to the TERMINAL tab, find the port, and click the globe icon.

You'll land on a `https://something-5000.app.github.dev` URL. That's your service.

Making a port public (Lab 5 backup): in the **PORTS** tab, right-click the port → **Port Visibility** → **Public**. Now anyone can reach that URL — useful if Hugging Face is blocked. Set it back to Private afterwards.

Stopping and resuming — protect your free hours

The free tier gives **60 core-hours per month**. A 2-core Codespace burns **2 hours of quota per hour** it is running. Our 15-hour workshop needs ~30 — comfortable, **as long as you stop it**.

At the end of each day:

- Go to **github.com/codespaces**
- Find your Codespace → click ... → **Stop codespace**

Next morning: same page → click your Codespace → it resumes in ~20 seconds with all your work intact. **Do not create a second one** — you'd lose your files and burn double the quota.

A Codespace also auto-stops after 30 minutes idle. Your files are safe; it just pauses.

Common mistakes

Mistake	What you'll see	Fix
Running <code>source .venv/bin/activate</code>	<code>No such file or directory</code>	Skip it. There's no venv in Codespaces
Opening <code>localhost:5000</code> in your own browser	Page won't load / connection refused	Use the PORTS tab or the Open in Browser pop-up
<code>mlflow ui</code> with no arguments	"No Experiments Exist", or a <code>MissingConfigException</code> traceback	Always: <code>mlflow ui --backend-store-uri sqlite:///mlflow.db --port 5000</code>
Deleting the <code>mlruns/</code> folder to "clean up"	Your logged models vanish; Register Model fails	Don't delete it — that folder holds your artifacts
Creating a new Codespace each day	Yesterday's work is missing	Resume the existing one at github.com/codespaces
Forgetting to stop it	Free hours drain overnight	Stop it at the end of each day
Running commands in the wrong folder	<code>file not found</code>	<code>cd /workspaces/mlops-sample-project</code> first
Closing the browser tab mid-lab	Nothing lost	Reopen from github.com/codespaces — it's still running

FAQs

Do I need to install anything at all? No. A browser and a verified GitHub account.

Is it really free? Yes — 60 core-hours and 15 GB of storage per month on the free plan. No card. Students can also get much larger quotas free via the **GitHub Student Developer Pack** (education.github.com/pack).

Will I lose my work when I close the tab? No. Files persist inside the Codespace. Closing the tab or stopping the Codespace keeps everything. Only **deleting** the Codespace removes your work — so push to GitHub on Day 3 (Lab 4) to keep a permanent copy.

Can I use my phone / tablet? It technically loads, but typing commands is painful. Use a laptop.

My Codespace is slow. Refresh the browser tab first. If it's still slow, stop and resume it from github.com/codespaces.

Can I use VS Code on my desktop instead of the browser? Yes — install the *GitHub Codespaces* extension and connect. Same environment, nicer editor. Do this only if you're already comfortable with VS Code.

Does Docker really work inside it? Yes. Docker-in-Docker is enabled, so Lab 3 works exactly as written. `docker run hello-world` proves it.

Does the campus network need anything special? It must be able to reach **github.com** and `*.app.github.dev`. If the campus filter blocks the second one, you can't reach your Codespace — tell the instructor immediately and tether to your phone.

I switched from laptop to Codespaces mid-workshop. Do I lose anything? No. The Codespace starts from the same repo. You'll redo a couple of quick commands, that's all.

What if it opens in "Recovery mode"? That means the container failed to build. First try **Ctrl/Cmd + Shift + P** → *Codespaces: Rebuild Container*. If that fails, delete the Codespace and create a fresh one. Tell the instructor — it's usually a one-line fix.

If something goes badly wrong

Try these in order:

1. **Reload** the browser tab.
2. **Stop and resume** at github.com/codespaces (⋮ → Stop, then reopen).
3. **Rebuild the container:** Ctrl/Cmd + Shift + P → *Codespaces: Rebuild Container*.
4. **Delete and recreate.** You lose uncommitted files, so `git push` first if you can. A fresh Codespace takes ~3 minutes.

Still stuck? Raise your hand — don't burn lab time debugging alone.